

# Physics Subjects

- Special Relativity
- Quantum Mechanics
- Classical Field Theory

The corresponding courses of lectures can be found at <https://teach-in.ru/>

# System of units

$$\hbar = c = 1$$

## Dimensions:

$$[\text{Energy}] = [\text{mass}] = [\text{length}]^{-1} = [\text{time}]^{-1}$$

## Relationships:

— Energy:  $E^2 = \mathbf{p}^2 + \mathbf{m}^2$   
 $E = \omega$

— Wavelength:  $\lambda = \frac{1}{p}$

— Time:  $t = \frac{1}{E}$

## Conversions:

$$\hbar c = 197.327 \text{ MeV fm}$$

$$1 \text{ GeV}^{-1} = 0.197 \text{ fm}$$

$$1 \text{ fm} = 5.07 \text{ GeV}^{-1}$$

$$1 \text{ s} = 1.52 \times 10^{24} \text{ GeV}^{-1}$$